MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER

TYPE 8767

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Geiger - Mueller, very high beta sensitivity, very thin mica window frisker probe, low background, halogen, self-quenching, ruggedized

See figure 1

Mounting position: Any

Weight: 6.5 ounces (184.3 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ebb	TA	Rp (external resistance)	Window thickness
Unit:	Vdc	°C	Meg	mgs/cm^2
Maximum:	950	+55	---	2.0
Minimum:	850	-20	3.3	1.4
TEST CONDITIONS:	900	---	3.3	---

GENERAL:

Qualification - Required

Marking (see note 2)

Holding period (MIL-STD-1311) - See note 3

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
6201	Background, contamination, and photosensitivity	4, 5	t = 5 minutes	0.65	II	N/t	---	35	Npm
6211	Starting voltage	-	Pulse amplitude = 0.25 V; use test circuit shown on figure 5; Nps = 200 (max)	0.65	II	Es	---	750	Vdc
6216	Relative plateau slope	1, 4	Npm = 5,000	0.65	II	---	---	---	---
6221	Response count rate and current (beta)	4, 7	t = 2 minutes	0.65	II	---	---	---	---
---	Dead time	-	Nps = 5,000; use test circuit shown on figure 5	0.65	II	td	---	100	µs
	Quality conformance inspection, part 2								
1041	Shock	6, 9	Hammer angle = 20 degrees	---	---	---	---	---	---
6226	Pulse amplitude	-	Use test circuit shown on figure 5	---	---	eo	13	100	v
	Quality conformance inspection, part 3								
---	Life test	-	Group D; Npm = 200 (min); t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
6201	Background, contamination and photosensitivity	-		---	---	N/t	---	60	Npm
6211	Starting voltage	-		---	---	ΔEs	---	±15	Vdc
6226	Pulse amplitude	-		---	---	eo	13	---	v
1031	Variable frequency vibration	6, 9, 11		---	---	---	---	---	---
6205	Leakage current	10, 11	Ebb = 500 Vdc	---	---	Lib	---	0.5	µAdc
---	Temperature cycling	8, 11		---	---	---	---	---	---

NOTES:

1. Npm - N_{850} = ±0.05 Npm maximum.
   Npm - N_{950} = ±0.05 Npm maximum.
   Npm = 5,000 cpm nominal at Vdc = 900 Vdc.
   N_{850} = Count rate at Vdc = 850 V.
   N_{950} = Count rate at Vdc = 950 V.
2. Each tube shall have an individual serial number which shall be legibly punched or engraved into the cathode.
3. The tube shall meet the initial acceptance requirements for background, contamination, and photosensitivity and starting voltage, and shall not deviate by more than 10 volts from the value at the beginning of the last 60-day period. A tube that fails the starting voltage test after the holding period, may be retested after an additional 20 or more days. Such a tube shall still meet the initial acceptance requirements and the starting voltage shall not deviate by more than 10 volts from the value at the beginning of the last 60-day period.

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NOTES: - Continued

4. Tube count rates shall be determined using a scaler having a maximum pulse pair resolving time of 10 µs and a pulse height discrimination level of 0.25 volt.

5. The tube end window shall be exposed to radiation from a General Electric 15-watt germicidal lamp and a General Electric 15-watt fluorescent lamp, or equivalents. Light sources shall be located not more than 1 inch from the mica window surface. (See figure 2.)

6. The criterion for passing this test shall be compliance of at least 80 percent of the tubes, after test, with the requirements for starting voltage and relative plateau slope.

7. The counting rate shall be not less than 10 percent below the value of the count rate specified in the EON TC-99 excitation unit, or equivalent. (See figure 3.)

8. With the tube in a field giving 100 ±10 Nps at 900 Vdc, determine tube response (count rate) at each of the following temperatures and in the order shown:

   (1) Room temperature.
   (2) -20°C.
   (3) Room temperature.
   (4) +55°C.
   (5) Room temperature.

   A minimum stabilization time of 30 minutes shall be allowed at each temperature. The absolute count rate at 900 Vdc shall not differ from the initial readings at room temperature by more than 10 percent at any of the four subsequent temperatures.

9. The tube shall be mounted by means of a rigid fixture which clamps the detector at the cathode as shown on figure 4.

10. Test shall be made at ambient conditions of normal room temperature and humidity.

Ⓐ 11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double plan shall be reinstated after three consecutive samples have been accepted.

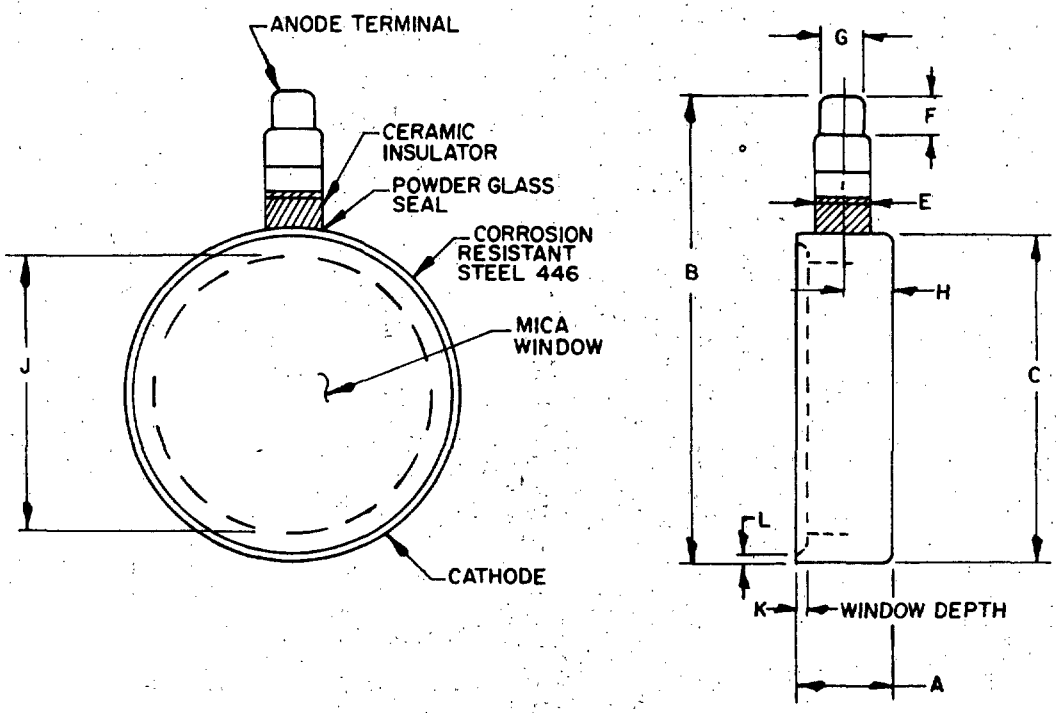
User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N051)

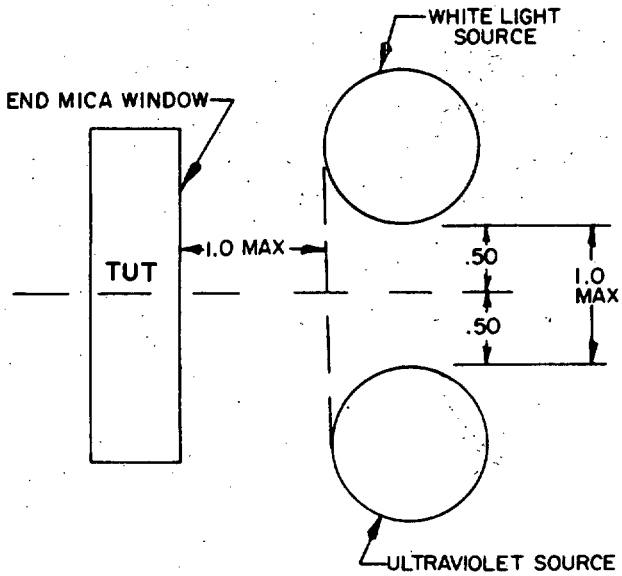
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	0.594 (15.09)	0.624 (15.85)
B		3.00 (76.20)
C	2.100 (53.34)	2.120 (53.85)
E	0.334 (8.48)	0.354 (8.99)
G	0.245 (6.22)	0.255 (6.48)
J		1.75 (4.45)
L	0.030 (.76)	0.040 (1.02)
Reference dimensions		
F	0.250 (6.35)	
H	0.313 (7.95)	
K	0.063 (1.60)	

Ⓐ FIGURE 1. Outline drawing of electron tube type 8767.

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NOTE: Axis of tube and centers of lamps lie in plane of paper.

FIGURE 2. Position of tube for photosensitivity test.

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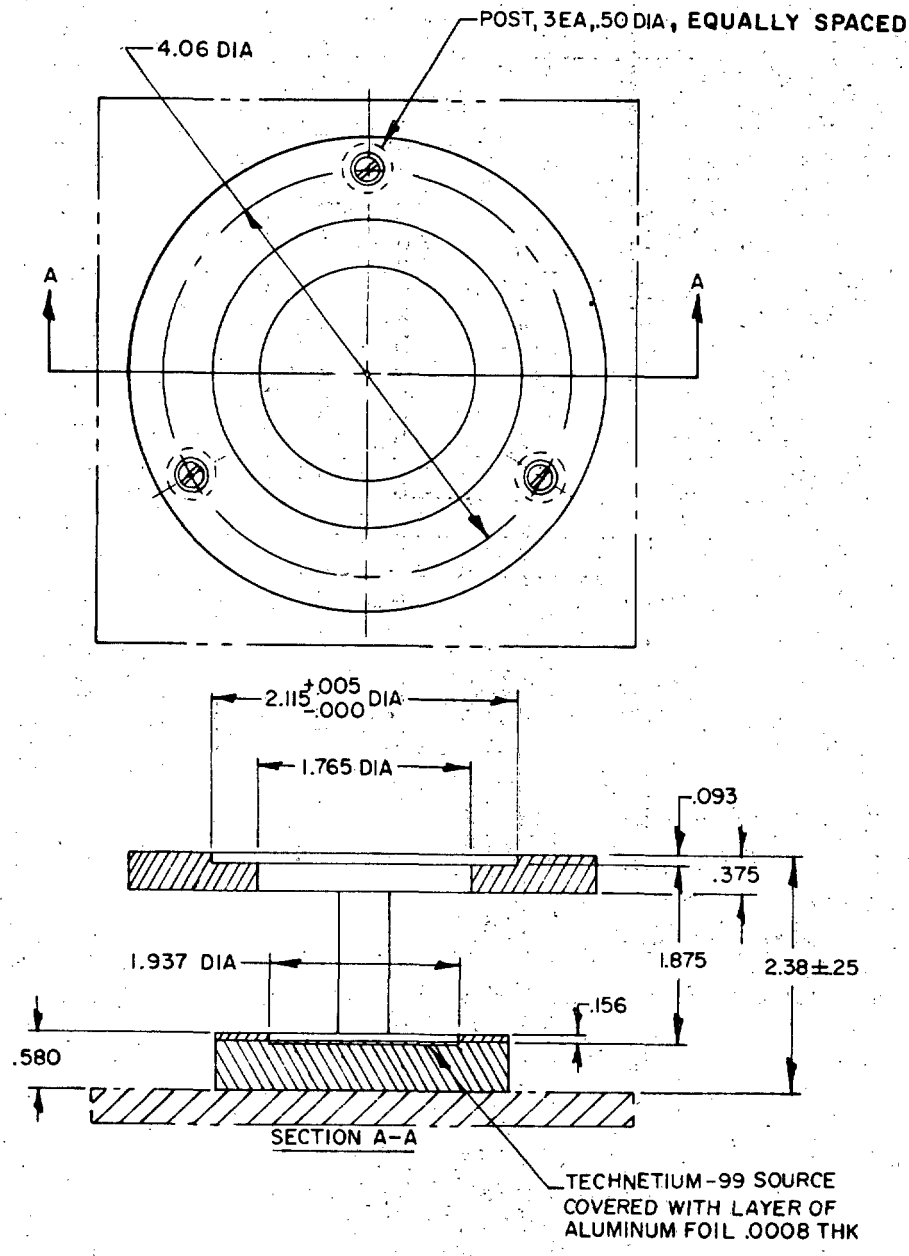
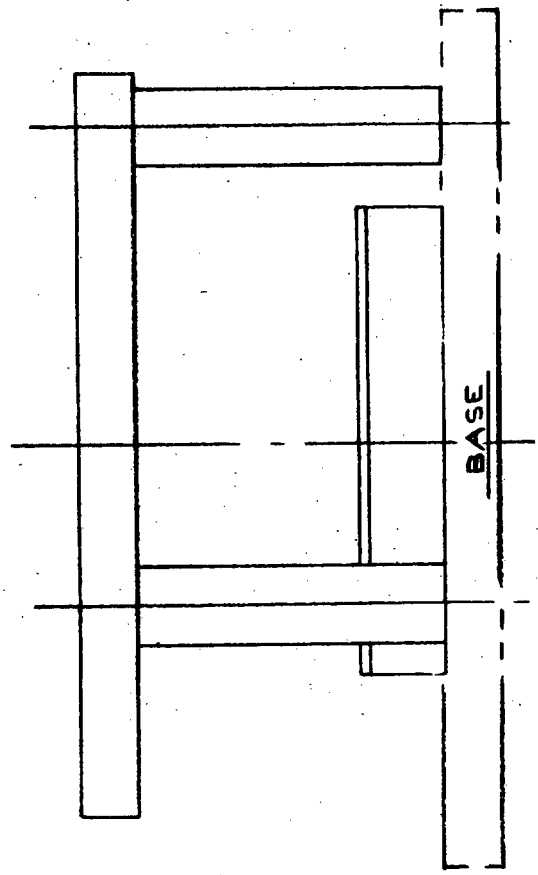


FIGURE 3. Excitation fixture.

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Beta fixture:

Bogey value for beta response is 7, 000 Npm.

FIGURE 3. Excitation fixture - Continued.

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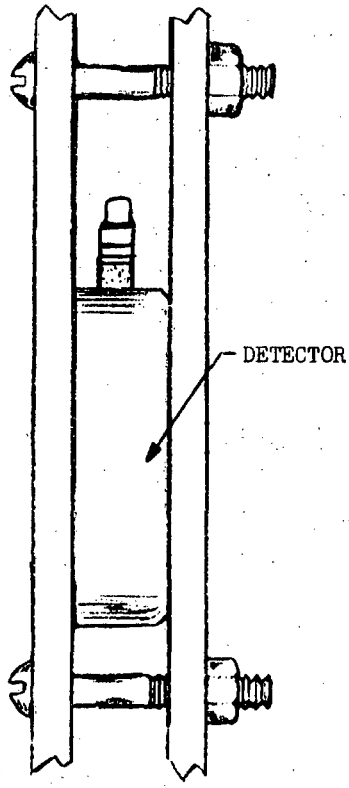


FIGURE 4. Method of clamping detector for shock and vibration tests.

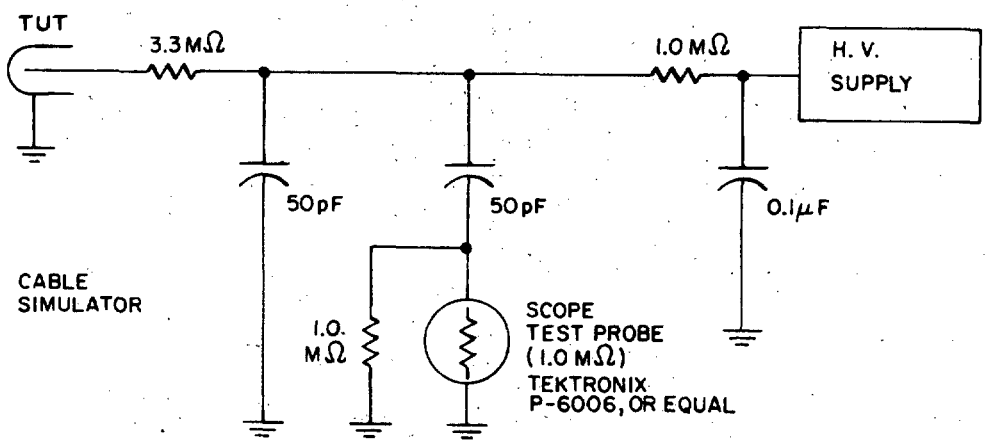


FIGURE 5. Test circuit.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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